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Breaking the sensitivity-frequency trade-off in a two-photon 3D-printed hollow-tympanic diaphragm Fabry–Perot fiber ultrasonic sensor

Jialin Wen,[1,2] Slawomir Ertman,[3] Dora Juanjuan Hu,[4] Jan Kośnic,[3] Linfeng Lan,[2] Tianyu Yang,[1,7] Yuming Dong,[1,8] Quandong Huang,[5] Perry Ping Shum,[6] Xinyong Dong,[5] Tomasz R. Wolinski,[3] and Huanhuan Liu[1,9]

[1]Center for Opto-Electronic Engineering and Technology, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China
[2]Guangdong Basic Research Center of Excellence for Energy & Information Polymer Materials, National Key Laboratory of Luminescent Materials and Devices, South China University of Technology, Guangzhou 510640, China
[3]Faculty of Physics, Warsaw University of Technology, PL-00662 Warsaw, Poland
[4]Institute for Infocomm Research, A*STAR, Singapore 138632, Singapore
[5]Institute of Advanced Photonics Technology, School of Information Engineering, Key Laboratory of Photonic Technology for Integrated Sensing and Communication, Guangdong University of Technology, Guangzhou 511400, China
[6]Department of Electrical and Electronic Engineering, Southern University of Science and Technology, Shenzhen 518055, China
[7]e-mail: ty.yang@siat.ac.cn
[8]e-mail: ym.dong@siat.ac.cn
[9]e-mail: hh.liu2@siat.ac.cn

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Developing ultra-sensitive miniature fiber sensors capable of detecting ultrasonic signals in the kilohertz range is of fundamental importance for a wide range of applications. However, current Fabry–Perot interferometric (FPI) fiber sensors face the main challenge of the intrinsic trade-off between sensitivity and resonant frequency, which hinders broadband ultrasonic detection. To break the sensitivity-frequency trade-off, we have proposed and demonstrated an FPI sensing diaphragm based on a hollow-tympanic diaphragm (HTD) structure. The HTD design integrates a centrally thinned tympanic-inspired region to enhance sensitivity and a peripheral hollow support to maintain the resonant frequency, effectively overcoming the trade-off between sensitivity and bandwidth. It is fabricated on the fiber end face using two-photon 3D printing technology for constructing an FPI acoustic fiber sensor. The obtained mechanical sensitivity reaches 1435 nm/kPa at the center frequency of 198 kHz, demonstrating that the HTD design effectively maintains high sensitivity within the high-frequency range. This result verifies that the proposed structure successfully mitigates the trade-off between the sensitivity and frequency response. Furthermore, the device successfully captures continuous pulsed ultrasonic signals generated by a spark discharge source, with demodulated results showing excellent agreement with the source frequency, confirming its potential for high-resolution acoustic field detection and transient ultrasonic measurement. Such a high-sensitivity probe capable of operating effectively in the high-frequency detection range holds great potential for applications in industrial ultrasonic monitoring, biomedical sensing, and structural health diagnostics.

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1. INTRODUCTION

Ultrasonic sensors play a vital role in fields such as medical imaging [1–3], nondestructive testing [4,5], structural health monitoring [6,7], and industrial surveillance [8]. At present, most applications still rely on traditional piezoelectric ultrasonic sensors. Although electrical sensors are technologically mature and widely used, their susceptibility to electromagnetic interference, bulky size, and limited capability for long-distance signal transmission make them unsuitable for operation under harsh conditions, thereby significantly restricting their application scope [9]. With the increasingly complex detection environment, the demand for miniaturization and the anti-interference ability of sensors is increasing. Traditional electrical sensors are unable to meet the requirements of the challenging conditions. Optical fiber ultrasonic sensors have shown significant advantages in complex detection environments due to their small size, immunity to electromagnetic interference, and low transmission loss [10,11]. The Fabry–Perot interferometric (FPI) sensor has become an ideal choice for detecting ultrasound in extreme environments due to its simple structure, high sensitivity, and batch manufacturing [12,13].

In ultrasound detection application scenarios, the sound pressure of ultrasound is typically low, so higher requirements are placed on the sensor's sensitivity. However, in the pursuit of high sensitivity for FPI sensors, a common challenge is the resulting low natural frequency. This is due to the inverse correlation between the first-order natural frequency of the diaphragm and the sensitivity [14]. The value of the natural frequency determines the frequency detection range of the sensor itself, so this poses a new challenge to the design of the FPI ultrasonic sensor [15].

In recent years, researchers have proposed various diaphragm structure optimization strategies to balance the relationship between natural frequency and sensitivity. Zhu *et al.* reported a novel FPI microphone based on an Au/PI composite film, which exhibited a flat response in the 200 Hz–10 kHz frequency range and improved the sensitivity to approximately 170 mV/Pa [16]. Zhang *et al.* proposed and verified a high-sensitivity cross-axial extrinsic FPI acoustic sensor. The cross-axial extrinsic FPI sensor is composed of the optical fiber and the inner surface of a cantilever, assisted with an adjustable 45° mirror, achieving a voltage sensitivity of 70.67 mV/Pa at the resonance frequency [17]. Tenbrake *et al.* designed a box-shaped fiber optic FPI sensor, in which an optomechanical spring tuning effect of the mechanical resonator frequency is observed at cryogenic temperatures, exceeding the resonator's linewidth and enhancing the integration capability of coupled resonators [18]. These methods improve the mechanical properties and acoustic response of FPI sensors to varying degrees by changing the material [19,20], structure [21,22], and parameters of the sensing diaphragm [23]. However, challenges such as high fabrication complexity, limited miniaturization, and only marginal improvement in specific performance indicators remain [24]. Therefore, for FPI sensors, it remains essential to explore diaphragm designs that can achieve high sensitivity in the high-frequency range while overcoming the inherent trade-off between the sensitivity and bandwidth.

In this work, we propose and demonstrate the hollow-tympanic diaphragm (HTD) as a novel diaphragm structure inspired by the tympanic membrane and enhanced by a hollowed peripheral support. The HTD consists of a centrally thinned region that mimics the tympanic membrane, enabling high displacement response and enhanced sensitivity, while a hollowed ring structure is introduced around the periphery to reduce damping and optimize stress distribution. The sensing probe is directly printed on the fiber end surface by two-photon 3D printing technology, which realizes the miniaturization of the device. By structuring the sensing diaphragm, the trend of natural frequency reduction is weakened, and the response of the sensor to the ultrasonic wave is significantly enhanced. The experimental results show that the proposed sensor can achieve a mechanical sensitivity of 1435 nm/kPa at the center frequency of 198 kHz, demonstrating that the HTD design effectively maintains high sensitivity within the high-frequency range. This work effectively balances the requirements of detection frequency and high sensitivity, and successfully addresses the issues of complex fabrication, limited miniaturization, and the performance ceiling of existing sensors. It provides a new feasible approach for the development of high-performance, miniaturized fiber optic ultrasonic sensors.

2. MECHANISM AND STRUCTURAL DESIGN

The extrinsic FPI resonant cavity is composed of a sensing diaphragm and a supporting structure. When light enters the resonant cavity from inside the optical fiber, the second-reflected light forms an interference beam with a stable phase. When the sensing diaphragm detects external pressure and undergoes deformation, the phase of the interference light changes, thereby mapping the variation in the measured parameter. Therefore, the FPI resonant cavity diaphragm is the key sensing element, and its size and the mechanical properties of its material determine the sensitivity and detection frequency range of the FPI sensing probe. At present, the most common sensing diaphragm is a circular diaphragm (CD) without any patterns. When an external force is applied uniformly to the diaphragm, the central displacement of the rigidly clamped CD is expressed as [25]

$$d = \frac{3(1-\nu^2)PR^4}{16EH^3}, \tag{1}$$

where ν and E are the Poisson's ratio and Young's modulus of the diaphragm material, respectively, P is the pressure applied to the diaphragm, and R and H are the radius and thickness of the diaphragm, respectively. The central deflection of the CD is only related to the material characteristics and size. The first-order natural frequency of CD is expressed as [25]

$$f = \frac{10.21H}{2\pi R^2}\sqrt{\frac{E}{12\rho(1-\nu^2)}}, \tag{2}$$

where ρ is the density of diaphragm material. It can be seen from the equation that changes in the diaphragm parameters lead to opposite trends in the diaphragm's natural frequency and central deflection. Therefore, the CD can only enhance sensitivity by reducing the frequency detection range, which makes it unsuitable for high-frequency signal detection and limits the application scenarios of extrinsic FPI sensors.

Therefore, it is necessary to design the structure of the sensing diaphragm so that structural modifications can maximize sensitivity while maintaining a high resonant frequency. In this work, hollow regions were introduced at the periphery of the conventional CD. In contrast, the thickness of the inner central region was reduced, with structural details shown in Fig. 1(a). Here, r is the inner radius, h is the thickness of the thinning, and L is the width of the cantilever beam. The hollow structure can reduce the damping effect generated when the diaphragm vibrates and decreases the overall stressed area, thereby increasing the displacement variation of the diaphragm. An internally thinned diaphragm with four hollowed sides is designed and fabricated on the fiber end face, as shown in Fig. 1(b). Meanwhile, the thickness reduction of the central region can

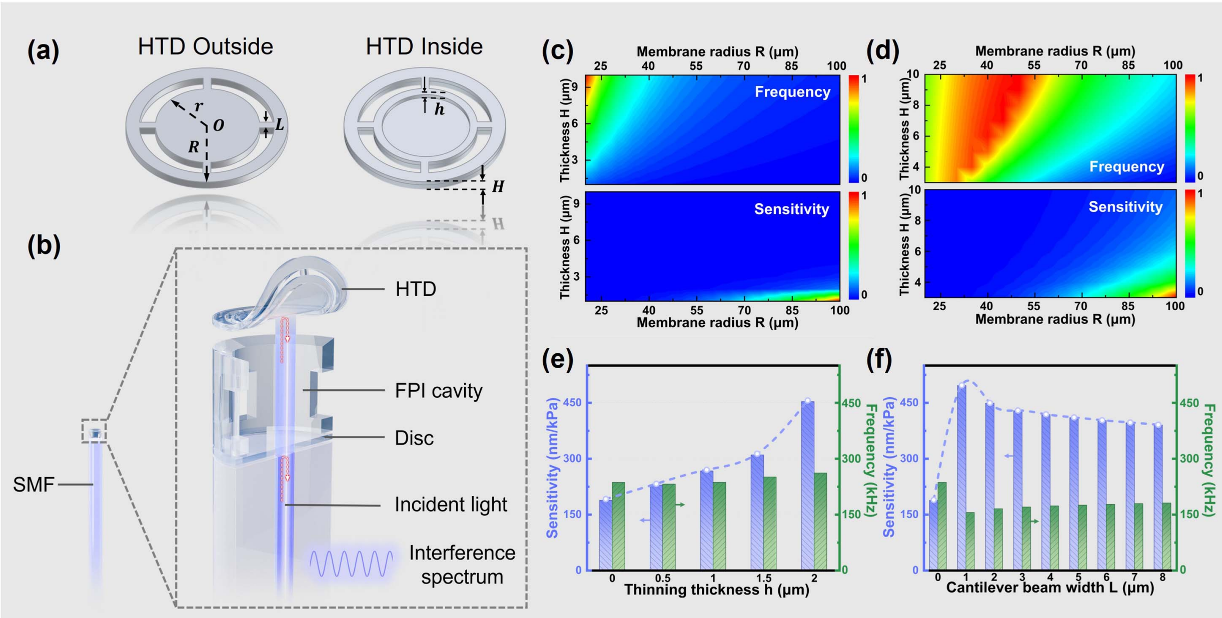

Fig. 1. (a) Schematic diagram of the HTD structure. (b) Schematic diagram of the HTD FPI sensor. Contour plot of the normalized sensitivity and natural frequency for the (c) conventional CD and (d) HTD. Comparison of the effects of (e) thinned region thickness and (f) cantilever beam width on diaphragm performance.

improve the responsivity of the diaphragm to the external pressure while maintaining the natural frequency, which broadens more application scenarios of the sensor. Finite element simulation is used to design and optimize diaphragm parameters, and a photoresist specialized for two-photon 3D printing is employed as the diaphragm material to perform simulation analyses on structures with different parameter configurations. Contour plots of the simulation results for both the CD and the HTD structures are shown in Figs. 1(c) and 1(d). The upper and lower panels correspond to the normalized natural frequency and sensitivity of the diaphragm under the same parameter conditions. It is evident that, for the CD, natural frequency and sensitivity exhibit opposite trends under identical conditions. In contrast, the HTD structure effectively alleviates the trade-off between natural frequency and sensitivity, achieving high sensitivity within a high-frequency range. The non-uniform distribution observed at the bottom boundary of the red region in Fig. 1(d) is mainly attributed to the discrete step size used in the parametric sweep and the nonlinear dependence of the natural frequency and sensitivity on the diaphragm radius and thickness [26]. During interpolation for contour plotting, such effects can lead to local fluctuations near the boundary.

Figures 1(e) and 1(f) illustrate the effect of the hollow structure and the thinned region on the overall diaphragm parameters. Figure 1(e) shows the relationship between sensitivity and the first-order natural frequency for diaphragms with only a thinned region (no hollow areas). The overall diaphragm size is kept constant while varying only the thickness of the central thinned region. It can be observed that, as the h increases, that is, reducing the remaining diaphragm thickness, the natural frequency of the diaphragm changes only slightly, whereas the central displacement increases rapidly. Figure 1(f) shows the influence of different cantilever beam widths L on the performance of the diaphragm when the central thinning structure remains unchanged. When L increases from 0 to 1 μm, a hollow supporting structure begins to form, breaking the continuous peripheral constraint. This significantly reduces the equivalent damping during vibration, allowing the external load to drive the diaphragm deformation more effectively, which leads to a rapid increase in the central displacement and a noticeable enhancement in sensitivity [27,28]. As L further increases, the cantilever region gradually becomes wider, and the hollow area correspondingly decreases, leading to a slight increase in the equivalent damping and stiffness during vibration and resulting in a minor reduction in sensitivity. However, since the hollow structure still occupies a relatively large proportion, the sensitivity remains significantly higher than that of the diaphragm without a hollow structure, indicating that the hollow design is overall beneficial for enhancing sensing performance. Therefore, this work combines the thinned area with the hollow structure to obtain a high-sensitivity sensing probe that satisfies the frequency requirements and is more suitable for detecting weak ultrasonic signals. For subsequent fabrication and measurement, h is fixed at 1 μm, and L is set at 5 μm. Meanwhile, the internal parameters of the structure are simulated and adjusted to determine the optimal diaphragm parameters for fabrication and subsequent experiments.

Figure 2(a) illustrates the schematic diagram of the HTD sensor, which consists of an optical fiber (single-mode fiber, SMF-28), a circular disk, a rectangular hollow supporting structure, and a sensing diaphragm. The thin circular disk structure with a thickness of 10 μm serves as the medium layer to connect the sensing probe and the fiber end face strongly. The operational mechanism can be explained by the opto-acoustic

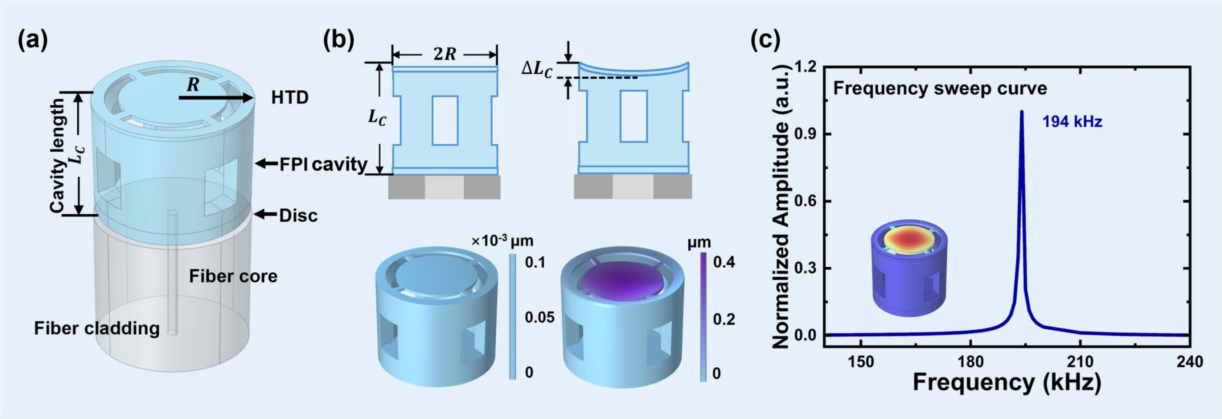

Fig. 2. Principle of the FPI ultrasonic sensing cavity. (a) Schematic diagram of the sensor based on the fiber end face, where the supporting structure and diaphragm collectively form an FPI resonant cavity. (b) Deformation of the FPI cavity under acoustic wave excitation and simulated deformation of the FPI cavity. (c) Modal frequency domain analysis simulation of the designed sensor device.

synergistic effect generated by the FPI cavity formed by the HTD. Acoustic vibrations induce a slight pressure acting on the sensing diaphragm, leading to deformation and thereby causing a change in the cavity length. Figure 2(b) shows the deformation of the sensing probe under external signals, where the diaphragm deformation modifies the FPI cavity length ΔL_C and produces corresponding variations in the interference signal.

To further investigate its mechanical properties, the finite element method (FEM) is used for simulation. A uniform pressure of 1 kPa is applied to the sensing diaphragm to observe its deformation, which is compared with the model without applied force. It is evident that the diaphragm exhibits significant deformation under pressure, demonstrating the sensor's responsiveness to external loading. The air domain environment is constructed to simulate and analyze the frequency response of the sensing probe. The center displacement of the sensor is scanned by frequency, as shown in Fig. 2(c). By taking into account the influence from the supporting ring of the FPI cavity, the first-order natural frequency of the probe is found to be 194 kHz when the effective radius of the HTD is taken as 57 μm, at which the maximum deformation occurred. The illustration of Fig. 2(c) shows the vibration mode of the sensor at the first-order resonance. Under uniform loading, the maximum displacement is located in the center of the diaphragm, satisfying the FPI cavity sensing requirements.

3. FABRICATION AND TEST

A. Device Fabrication and Characterization

Based on the simulation results, the FPI ultrasonic sensing cavity was fabricated at the top of SMF-28 fiber using the direct laser writing technique through two-photon polymerization [29–31]. The structure was fabricated in Warsaw Univ. of Technology (Poland) with a commercial system, the Photonics Professional GT2 (Nanoscribe GmbH). The printing material employed was IP-S resin, developed and supplied by Nanoscribe. According to the supplier's data, the polymerized material exhibits a Young's modulus of 2.1 GPa and an estimated Poisson's ratio of 0.3. The structure was printed using a 25× objective, with both slicing and hatching distances set to 0.3 μm, which allowed for sufficient fabrication detail and surface smoothness. After printing, the structure was developed in PGMEA and Novec to remove resin from the unpolymerized regions.

Figure 3(a) presents the overall morphology of the sensor, showing that the device is completely and smoothly fabricated on the fiber end face as observed by scanning electron microscopy (SEM). Figure 3(b) shows a three-quarter SEM view of the sensor, providing a clearer visualization of the internal supporting structure. It can be observed that the supporting structure is hollow inside. Figure 3(c) shows the measured reflected spectrum with the free spectral range (FSR) of 14.7 nm. The corresponding cavity length of the FPI sensor is calculated to be 81.7 μm, which is in good agreement with our designed cavity length of 82 μm. The disk film induced interference can modulate the spectral envelope of the main-cavity FPI interference spectrum. However, this fully solid-state disk is located at the innermost part of the sensing head. Therefore, the acoustic-wave-induced variations coupled to it are weaker than those of the outermost HTD film.

B. Ultrasonic Sensing

To evaluate the sensing performance of the HTD sensor, an intensity demodulation method is employed to detect the interference signal inside the cavity, and an ultrasonic detection system is established as shown in Fig. 4. A function generator (33600A, Keysight) is used to drive an ultrasonic transducer to generate ultrasonic signals, while a tunable laser source (TSL-570, Santec) serves as the optical source. The output wavelength of the tunable laser is fixed at the quadrature point of the sensor to obtain the maximum signal response. After passing through a circulator into the sensor, the light is reflected within the cavity, and the reflected interference light is converted into an electrical signal by a photodetector (DET01CFC/M, Thorlabs), which is then measured by a digital oscilloscope (EXR104A, Keysight).

The performance of the sensing probe is investigated. As shown in Fig. 5(a), the HTD sensor achieves a signal-to-noise

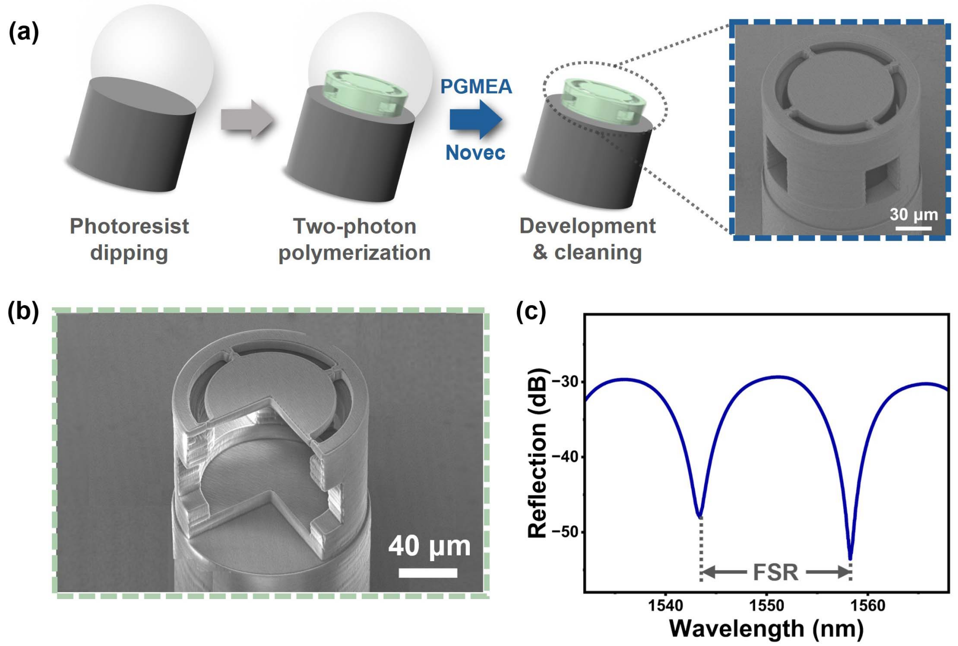

Fig. 3. (a) Process of two-photon 3D printing technology, including droplet deposition, two-photon printing, development, and sample cleaning. The SEM image shows the fabricated structure. (b) Sensor probe with 1/4 of the structure removed for clearer observation of the internal cavity. (c) Reflection spectrum of the sensor in air.

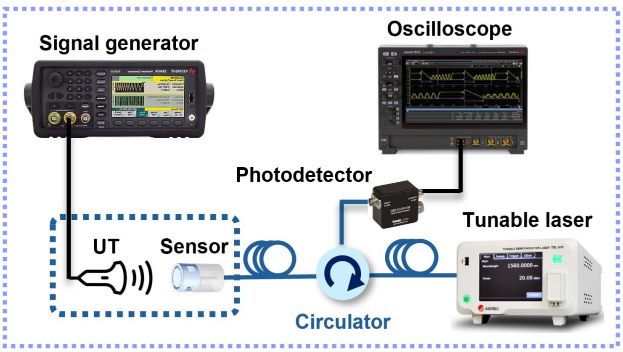

Fig. 4. Experimental optical path connection diagram (UT: ultrasonic transducer).

ratio (SNR) as high as 82.02 dB at its resonant frequency, enabling highly sensitive detection of weak signals at the corresponding frequency. The noise floor in the frequency domain is approximately −110 dB, which is comparable to the noise floor of the previous reports [32,33], indicating that it has detection ability. Generally, the noise mainly originates from unavoidable external and system-related factors. The noise floor can be further mitigated by a low noise light source [34], but it usually requires a higher cost. This value surpasses that of currently reported fiber optic ultrasonic sensors, demonstrating excellent signal performance.

In the experiment, the weak ultrasonic signal is generated by the ultrasonic transducer, and the ultrasonic transducer is connected to a signal generator to obtain a stable ultrasonic signal. Meanwhile, based on the SNR level, the minimum detectable pressure (MDP) of the sensor is calculated, which represents the smallest acoustic pressure the sensor can detect at this frequency. The calculation formula of the MDP is [35]

$$\mathrm{MDP} = \frac{P \times 10^{-\frac{\mathrm{SNR}}{20}}}{\sqrt{\Delta f}}, \tag{3}$$

where P is sound pressure as 0.6295 Pa [36], and Δf is the fast Fourier transform (FFT) frequency domain minimum interval, with a value of 5 kHz measured by oscilloscope. According to Eq. (3), the MDP of the sensor at its resonant frequency is calculated to be 0.71 $\mu\mathrm{Pa/Hz}^{1/2}$, with a noise floor of −110 dB. It indicates that, under resonant conditions, the sensor can detect extremely weak acoustic signals, demonstrating excellent ultrasonic detection performance, which is far superior to that of currently reported fiber optic ultrasonic sensors [37].

Figure 5(b) shows the frequency domain response range of the HTD sensor. The sound pressure is fixed, and only the ultrasonic frequency emitted by the ultrasonic transducer is changed. The SNR levels detected by the sensor at different frequencies are measured. By connecting the SNR values corresponding to different frequencies, the sensor's frequency response range can be obtained. The plot clearly shows that the highest SNR occurs at 198 kHz, indicating that the actual first-order natural frequency is 198 kHz. The slight frequency deviation from the simulation might be originally from material properties caused by curing conditions and environmental factors [38]. Accordingly, the amplitude-frequency response indicates a stable and pronounced sensitivity over 100–300 kHz, confirming the broadband detection capability of the sensor in this frequency band.

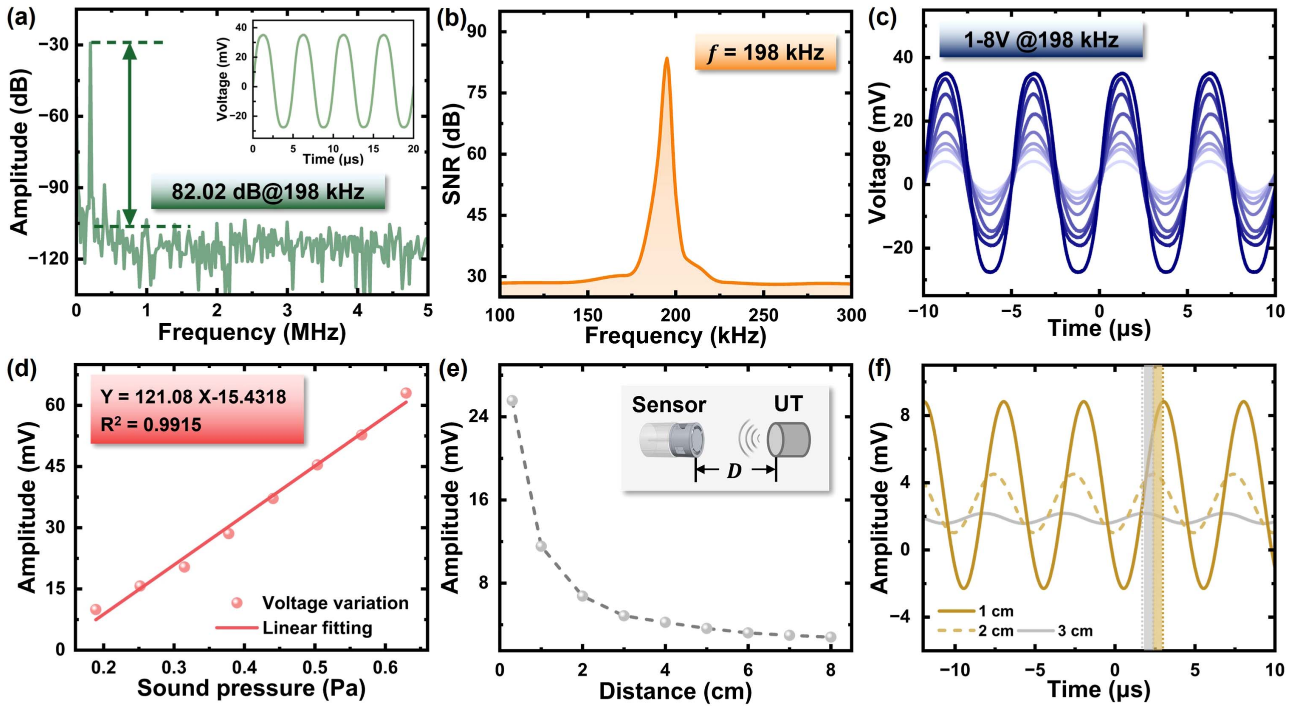

Fig. 5. Response characteristics of the HTD sensor. (a) SNR of the sensor for a 198 kHz sinusoidal signal. The inset shows the time-domain spectrogram detected by the sensor. (b) Frequency response curve of the sensor. (c) Time-domain signals collected by the HTD sensor at different voltages under the resonance frequency. (d) Voltage sensitivity of the HTD sensor. (e) The sensor measures the time signal emitted by the ultrasonic transducer at different distances. (f) Distance response of the HTD sensor.

Figures 5(c) and 5(d) show the response of the sensor to different sound pressures. The frequency emitted by the ultrasonic transducer is fixed (at the first-order natural frequency), and the pressure of the signal emitted by the signal generator is changed, corresponding to the sound pressure emitted by the ultrasonic transducer. Figure 5(c) shows the time-domain waveforms detected by the sensor under different sound pressures. It is evident that, as the sound pressure changes, the amplitude of the acoustic signal detected by the sensor changes accordingly. The amplitude of the time-domain curve corresponding to different sound pressure values is drawn into a point map. There is a linear relationship between the amplitude of the output signal detected and the input sound pressure, with a slope of 121.08 mV/V_{pp} and a linear correlation value of 0.99, as shown in Fig. 5(d). The mechanical sensitivity of the sensor is calculated to be 1435 nm/kPa.

The distance response of the HTD sensor is also tested, as shown in Figs. 5(e) and 5(f). The distance between the sound source and the sensor increases, and the amplitude of the detected acoustic signal gradually decreases, following an exponential decay trend. The illustration of Fig. 5(e) is the distance detection experimental device. Since the sound pressure of ultrasound in air is known to decay exponentially with distance, the observed decay trend in the sensor's detected signal indicates that the sensor can accurately detect acoustic signals in air at different distances. Figure 5(f) shows the time-domain waveforms detected by the sensor at different distances. As the detection distance increases, not only does the amplitude decrease, but a fixed phase shift also appears, confirming the sensor's ability to respond to sound pressure at varying distances.

Table 1 compares the performance of the HTD sensor with previously reported fiber optic ultrasonic sensors [14,19,35, 39–41], while Fig. 6 illustrates the relationship between frequency and sensitivity. It can be clearly observed that, although various types of sensors have been proposed in recent years,

Table 1. Comparison of Characteristics of the HTD Sensor with Reported Schemes

Type	Diameter (μm)	Thickness (μm)	Frequency (kHz)	Sensitivity (nm/kPa)	Ref.
FPI with supporting beam silicon diaphragm	1120	5	60	807	[39]
FPI with silicon dioxide clover diaphragm	600	10	135.84	71.17	[14]
FPI with corrugated membrane structure	120	7	100	367.73	[35]
FPI with multilayer graphene diaphragm	145	0.1	10	1100	[40]
FPI with a gold-plated silicon diaphragm	1120	5	60	733	[41]
FPI with spirally suspended structure in SU-8	120	–	5	182	[19]
FPI with HTD diaphragm	**120**	**3**	**198**	**1435**	**This work**

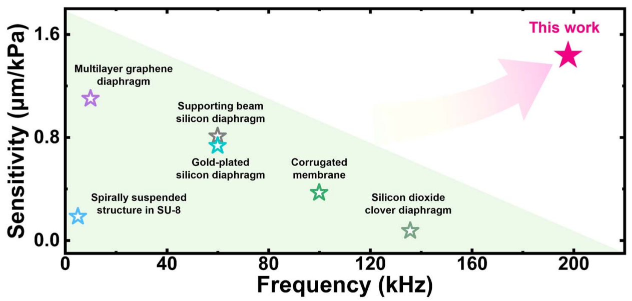

Fig. 6. The performance comparison of the HTD sensor with other reported schemes.

most of them are still constrained by the trade-off between frequency and sensitivity. Compared with previously reported designs, the intrinsic advantage of the HTD diaphragm is enhancing sensitivity without sacrificing the natural frequency. On one hand, the center-thinned region dominates the displacement response under external acoustic pressure, thereby significantly improving the sensitivity. On the other hand, the peripheral hollow structure preserves the overall effective stiffness and characteristic dimensions of the diaphragm and prevents excessive stiffness reduction, thus effectively suppressing a pronounced decrease in the natural frequency. This design decouples sensitivity from the natural frequency, fundamentally breaking the long-standing sensitivity-frequency trade-off and leading to a marked performance improvement. Moreover, because of its compact size, the sensor can be applied in more microscopic scenarios, further broadening the application scope of fiber optic ultrasonic sensing.

C. HTD Sensor for Ultrasonic Pulse Signal

Pulse signal detection plays a crucial role in industrial, power, and safety monitoring applications. In power systems, the detection of partial discharge pulses is an important means of assessing the insulation condition of high voltage equipment [42]. In structural health monitoring, impact pulses can be used to identify material cracks or fatigue damage [43]. Accurate detection of ultrasonic pulse signals is helpful for troubleshooting and provides an important guarantee for the maintenance and operation safety of industrial equipment. Therefore, the sensor for detecting ultrasonic pulse signals is required to have both a wide bandwidth and a high sensitivity.

In the experimental system, the sound source device is replaced by a pulse signal generator, as shown in Fig. 7(a). The HTD sensor directly detects the released pulse signal and observes the output on the oscilloscope. Figure 7(b) illustrates a uniform and continuous pulse signal appearing in the time-domain diagram detected by the sensor. The obtained continuous pulse signal is analyzed by FFT to obtain the normalized spectrum diagram shown in Fig. 7(c). It can be observed that two main frequency components appear in the spectrum, approximately 217.6 and 235.3 kHz, which correspond to the dominant frequencies of different individual pulses in the time-domain signal. Meanwhile, some weaker frequency components are also present, but their overall amplitudes are relatively small. This result demonstrates that the sensor can not only accurately detect the dominant frequencies of pulse signals, but also exhibits a good response over a wide frequency range, thereby enabling effective detection of weak ultrasonic pulse signals. Subsequently, the ability of the sensor to detect pulsed ultrasonic signals in extreme environments can be further verified, and its stability and reliability can be explored to provide a solid guarantee for equipment safety monitoring. Previous reports have confirmed that the optical fiber end face device prepared by two-photon polymerization has good stability and application value [44,45]. To evaluate the reliability of our developed FPI sensor, we have repeated experiments over 7 days. The optical spectrum of the FPI sensor has shown almost no degradation in amplitude and FSR, indicating the mechanical stability of the sensor during testing.

4. CONCLUSION

This work proposes a high-performance and miniaturized photoacoustic sensor design and fabrication method for the detection of pulsed ultrasonic signals. Specifically, a hollow-tympanic diaphragm FPI cavity is stably fabricated on the end face of a single-mode fiber using two-photon polymerization. The sensor achieves a mechanical sensitivity of 1435 nm/kPa at 198 kHz, effectively overcoming the typical trade-off between sensitivity and detection frequency. Furthermore, the HTD sensor is employed for pulse signal detection, enabling precise

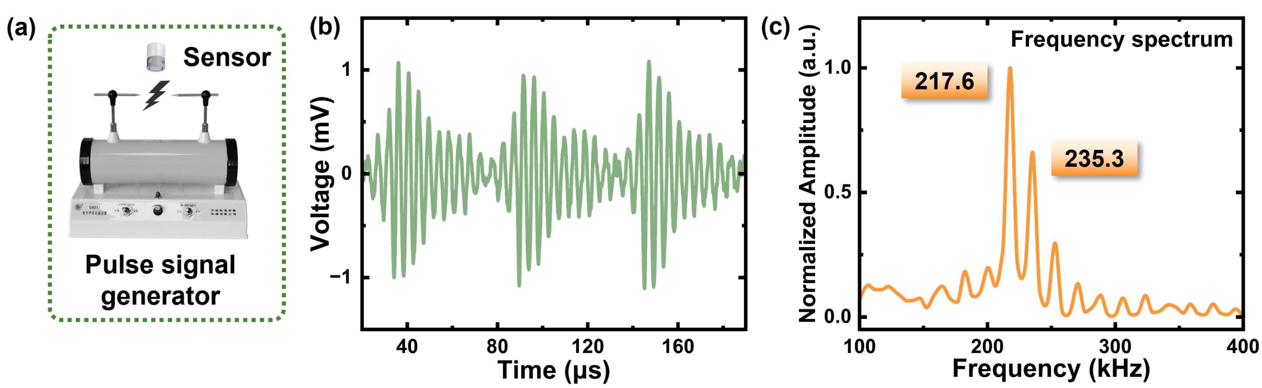

Fig. 7. (a) Experimental setup for detecting pulse signal. (b) Time-domain signal of the pulse signal detected by the sensor. (c) FFT of the time-domain pulse signal.

measurement and demodulation of the acquired signals. Overall, this work provides a feasible solution for weak acoustic signal detection and overcomes the limitations of conventional electrical acoustic sensors in extreme environments. More importantly, the sensor exhibits excellent adaptability and flexibility, demonstrating broad application potential across diverse scenarios.

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Data Availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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